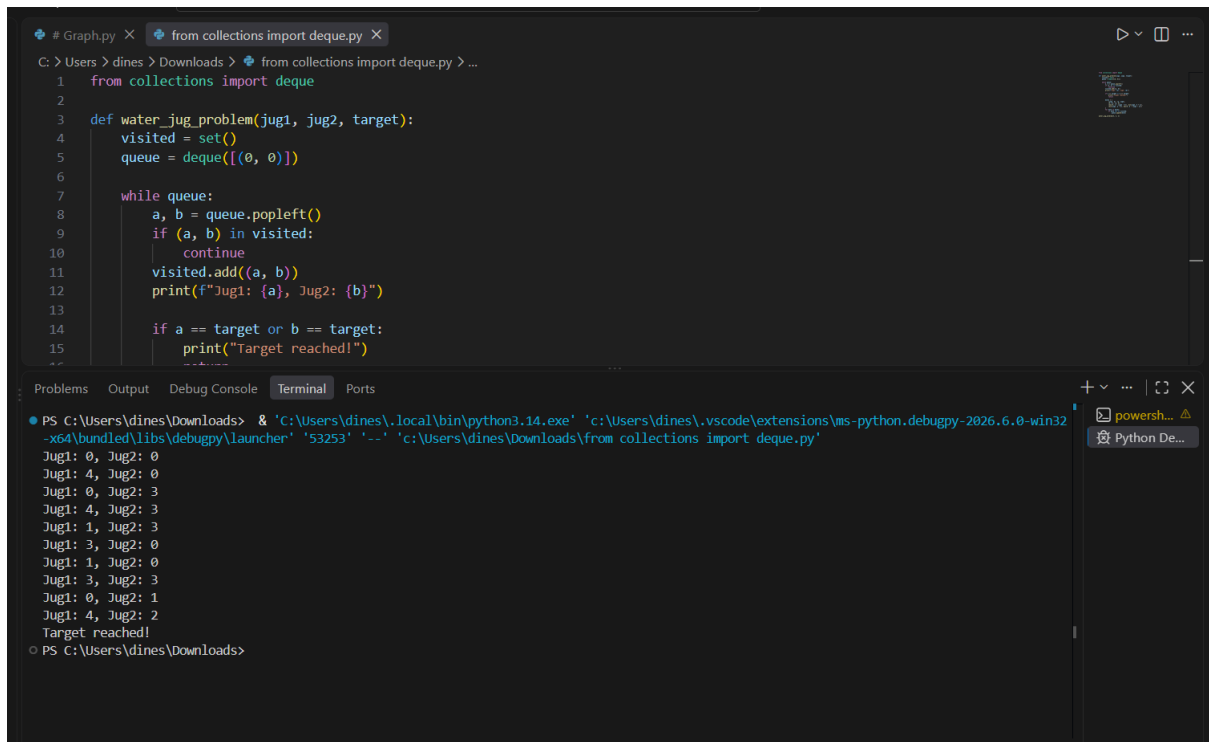


BAIRAGANI DINESH -192372189

Exp 1:

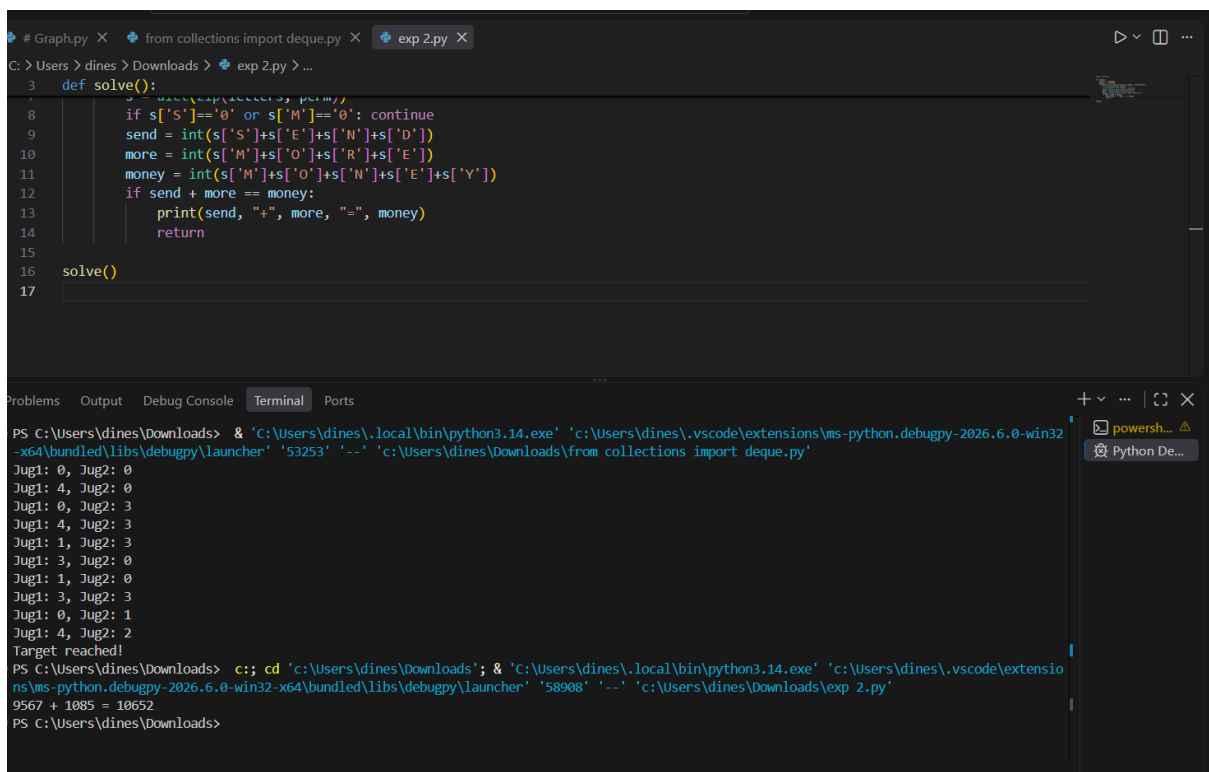


```
# Graph.py X from collections import deque.py X
C:\Users\dines> Downloads > from collections import deque.py > ...
1 from collections import deque
2
3 def water_jug_problem(jug1, jug2, target):
4     visited = set()
5     queue = deque([(0, 0)])
6
7     while queue:
8         a, b = queue.popleft()
9         if (a, b) in visited:
10             continue
11         visited.add((a, b))
12         print(f"Jug1: {a}, Jug2: {b}")
13
14         if a == target or b == target:
15             print("Target reached!")
16             return
17
18     return False
19
20 if __name__ == '__main__':
21     jug1 = 4
22     jug2 = 3
23     target = 2
24     water_jug_problem(jug1, jug2, target)
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Problems Output Debug Console Terminal Ports

```
PS C:\Users\dines\Downloads> & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '53253' '--' 'c:\Users\dines\Downloads\from collections import deque.py'
Jug1: 0, Jug2: 0
Jug1: 4, Jug2: 0
Jug1: 0, Jug2: 3
Jug1: 4, Jug2: 3
Jug1: 1, Jug2: 3
Jug1: 3, Jug2: 0
Jug1: 1, Jug2: 0
Jug1: 3, Jug2: 3
Jug1: 0, Jug2: 1
Jug1: 4, Jug2: 2
Target reached!
PS C:\Users\dines\Downloads>
```

Exp 2:



```
# Graph.py X from collections import deque.py X exp 2.py X
C:\Users\dines> Downloads > exp 2.py > ...
3 def solve():
4     # ... (code for solving the word puzzle) ...
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6     # ... (code for solving the word puzzle) ...
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8     # ... (code for solving the word puzzle) ...
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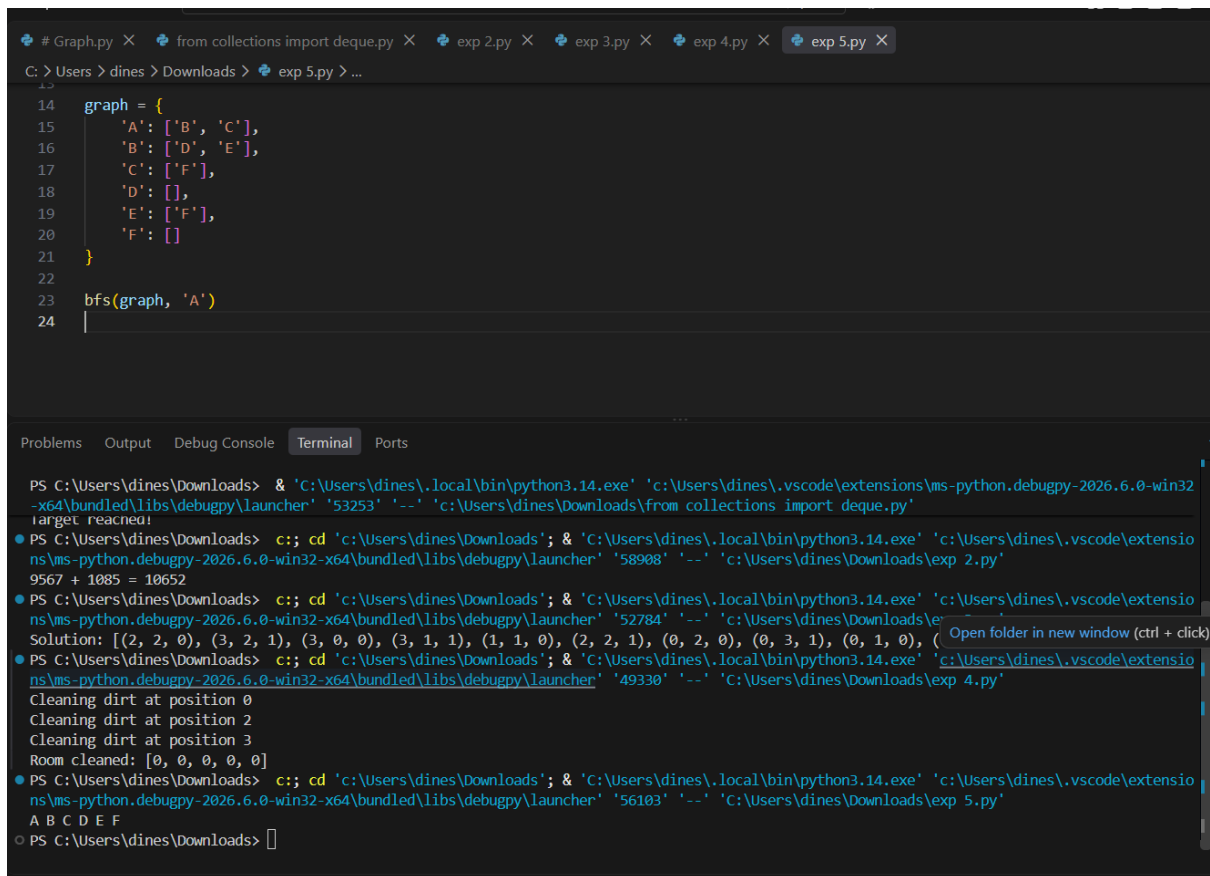
Problems Output Debug Console Terminal Ports

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PS C:\Users\dines\Downloads> & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '53253' '--' 'c:\Users\dines\Downloads\from collections import deque.py'
Jug1: 0, Jug2: 0
Jug1: 4, Jug2: 0
Jug1: 0, Jug2: 3
Jug1: 4, Jug2: 3
Jug1: 1, Jug2: 3
Jug1: 3, Jug2: 0
Jug1: 1, Jug2: 0
Jug1: 3, Jug2: 3
Jug1: 0, Jug2: 1
Jug1: 4, Jug2: 2
Target reached!
PS C:\Users\dines\Downloads> c;; cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '58908' '--' 'c:\Users\dines\Downloads\exp 2.py'
9567 + 1085 = 10652
PS C:\Users\dines\Downloads>
```

Exp 3:

```
# Graph.py X from collections import deque.py X exp 2.py X exp 3.py X
C: > Users > dines > Downloads > exp 3.py > ...
1 def missionaries_cannibals():
2     start = (3,3,1)
3     goal = (0,0,0)
4     visited = set()
5     stack = [(start, [])]
6
7     def valid(m,c):
8         return (m>=0 and c>=0 and m<=3 and c<=3 and (m==0 or m>=c))
9
10    while stack:
11        (m,c,b), path = stack.pop()
12        if (m,c,b) == goal:
13            print("Solution:", path)
14            return
15        if (m,c,b) in visited: continue
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```

Exp 5:-



The screenshot shows a VS Code editor with a Python script named `exp 5.py` and its execution output in the terminal. The script defines a graph and runs a breadth-first search (bfs) algorithm starting from node 'A'.

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

bfs(graph, 'A')
```

The terminal output shows the execution of the script, including the path to the Python interpreter and the output of the bfs function.

```
PS C:\Users\dines\Downloads> & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '53253' '--' 'c:\Users\dines\Downloads\from collections import deque.py'
target reached!
PS C:\Users\dines\Downloads> c:; cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '58908' '--' 'c:\Users\dines\Downloads\exp 2.py'
9567 + 1085 = 10652
PS C:\Users\dines\Downloads> c:; cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '52784' '--' 'c:\Users\dines\Downloads\exp 3.py'
Solution: [(2, 2, 0), (3, 2, 1), (3, 0, 0), (3, 1, 1), (1, 1, 0), (2, 2, 1), (0, 2, 0), (0, 3, 1), (0, 1, 0), (0, 0, 0)]
PS C:\Users\dines\Downloads> c:; cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '49330' '--' 'c:\Users\dines\Downloads\exp 4.py'
Cleaning dirt at position 0
Cleaning dirt at position 2
Cleaning dirt at position 3
Room cleaned: [0, 0, 0, 0, 0]
PS C:\Users\dines\Downloads> c:; cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '56103' '--' 'c:\Users\dines\Downloads\exp 5.py'
A B C D E F
PS C:\Users\dines\Downloads>
```

Exp 6:

```
# Graph.py
from collections import deque
exp 2.py
exp 3.py
exp 4.py
exp 5.py
exp 6.py

C: > Users > dines > Downloads > exp 6.py > ...
1 def dfs(graph, start, visited=None):
2     if visited is None:
3         visited = set()
4         visited.add(start)
5         print(start, end=" ")
6         for neighbor in graph[start]:
7             if neighbor not in visited:
8                 dfs(graph, neighbor, visited)
9
10    graph = {
11        'A': ['B', 'C'],
12        'B': ['D', 'E'],
13        'C': ['F'],
14        'D': [],
15        'E': ['F'],
16        'F': []
17    }
18
19    dfs(graph, 'A')
20
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\dines\Downloads> c:: cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '49330' '--' 'c:\Users\dines\Downloads\exp 4.py'
ns\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '56103' '--' 'c:\Users\dines\Downloads\exp 5.py'
A B C D E F
PS C:\Users\dines\Downloads> c:: cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '49699' '--' 'c:\Users\dines\Downloads\exp 6.py'
A B D E F C
PS C:\Users\dines\Downloads>
```

Exp 7:-

```
Users > dines > Downloads > exp 7.py > ...
3 def tsp(graph, start):
4     cost = 0
5     k = start
6     for j in perm:
7         cost += graph[k][j]
8         k = j
9     cost += graph[k][start]
10    if cost < min_cost:
11        min_cost = cost
12        min_path = (start,) + perm + (start,)
13    print("Path:", min_path, "Cost:", min_cost)
14
15    graph = {
16        'A': {'B':10, 'C':15, 'D':20},
17        'B': {'A':10, 'C':35, 'D':25},
18        'C': {'A':15, 'B':35, 'D':30},
19        'D': {'A':20, 'B':25, 'C':30}
20    }
21    tsp(graph, 'A')
22
```

Problems Output Debug Console Terminal Ports

```
C:\Users\dines\Downloads> c:: cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '63940' '--' 'c:\Users\dines\Downloads\exp 5.py'
ns\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '63940' '--' 'c:\Users\dines\Downloads\exp 6.py'
B D E F C
C:\Users\dines\Downloads> c:: cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '63940' '--' 'c:\Users\dines\Downloads\exp 7.py'
Path: ('A', 'B', 'D', 'C', 'A') Cost: 80
C:\Users\dines\Downloads>
```

Exp 8:

```
# Graph.py X from collections import deque.py X exp 2.py X exp 3.py X exp 4.py X exp 5.py X exp 6.py X exp 7.py X exp 8.py X
C: > Users > dines > Downloads > exp 8.py > ...
3 def astar(start, goal, graph, h):
14     node = parent[node]
15     return path[::-1]
16     for neigh, cost in graph[node]:
17         new_g = g[node]+cost
18         if neigh not in g or new_g < g[neigh]:
19             g[neigh] = new_g
20             f = new_g + h[neigh]
21             heappush(open_list,(f,neigh))
22             parent[neigh]=node
23
24 graph = {
25     'A': [('B',1),('C',3)],
26     'B': [('D',1)],
27     'C': [('D',1)],
28     'D': []
29 }
30 h = {'A':3,'B':1,'C':1,'D':0}
31 print(astar('A','D',graph,h))
32
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\dines\Downloads> c;; cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '49699' '--' 'c:\Users\dines\Downloads\exp 6.py'
ns\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '63940' '--' 'c:\Users\dines\Downloads\exp 7.py'
Path: ('A', 'B', 'D', 'C', 'A') Cost: 80
● PS C:\Users\dines\Downloads> c;; cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '63909' '--' 'c:\Users\dines\Downloads\exp 8.py'
['A', 'B', 'D']
○ PS C:\Users\dines\Downloads> 
```

Exp 9:

```
C: > Users > dines > Downloads > exp 9.py > ...
1 import itertools
2
3 colors = ['Red','Green','Blue']
4 regions = ['A','B','C']
5 neighbors = {'A': ['B','C'], 'B': ['A','C'], 'C': ['A','B']}
6
7 def valid(assign):
8     for r in regions:
9         for n in neighbors[r]:
10             if assign[r]==assign[n]:
11                 return False
12     return True
13
14 for perm in itertools.product(colors, repeat=len(regions)):
15     assign = dict(zip(regions,perm))
16     if valid(assign):
17         print(assign)
18         break
19
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\dines\Downloads> c;; cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '63940' '--' 'c:\Users\dines\Downloads\exp 7.py'
ns\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '63909' '--' 'c:\Users\dines\Downloads\exp 8.py'
['A', 'B', 'D']
● PS C:\Users\dines\Downloads> c;; cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '60513' '--' 'c:\Users\dines\Downloads\exp 9.py'
{'A': 'Red', 'B': 'Green', 'C': 'Blue'}
○ PS C:\Users\dines\Downloads> 
```

Exp 10:-

```
ns import deque.py X exp 2.py X exp 3.py X exp 4.py X exp 5.py X exp 6.py X exp 7.py X exp 8.py X exp 9.py X exp 10.py X
C: > Users > dines > Downloads > exp 10.py > ...
1 def print_board(board):
2     for row in board:
3         print(" | ".join(row))
4         print("-"*5)
5
6 def check_winner(board, player):
7     for row in board:
8         if all(s==player for s in row): return True
9     for col in range(3):
10        if all(board[row][col]==player for row in range(3)): return True
11    if all(board[i][i]==player for i in range(3)): return True
12    if all(board[i][2-i]==player for i in range(3)): return True
13    return False
14
15 board = [[" "]*3 for _ in range(3)]
16 board[0][0]="X"; board[1][1]="X"; board[2][2]="X"
17 print_board(board)
18 print("winner?", check_winner(board, "X"))
19

Problems Output Debug Console Terminal Ports
PS C:\Users\dines\Downloads> c;; cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscod
ns\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '60555' '--' 'c:\Users\dines\Downloads\exp_10.py'
| X |
-----
| | X
-----
winner? True
PS C:\Users\dines\Downloads>

Navigate to Command
Scroll to Previous Command (Ctrl+UpArrow)
Scroll to Next Command (Ctrl+DownArrow)
Spaces: 4 UTF-8 Python Python 3.14.3.final.0
```

exp 11:-

```
exp 2.py X exp 3.py X exp 4.py X exp 5.py X exp 6.py X exp 7.py X exp 8.py X exp 9.py X exp 10.py X exp 11.py X
Users > dines > Downloads > exp 11.py > ...
def minimax(depth, nodeIndex, maximizingPlayer, values, alpha, beta):
    for i in range(2):
        val = minimax(depth+1, nodeIndex*2+i, False, values, alpha, beta)
        best = max(best, val)
    (function) def print(
        *values: object,
        sep: str | None = " ",
        end: str | None = "\n",
        file: SupportsWrite[str] | None = None,
        flush: Literal[False] = False
    ) -> None
    Prints the values to a stream, or to sys.stdout by default.
    sep
        string inserted between values, default a space.
    end
    print("Optimal value:", minimax(0,0,True,values,float('-inf'),float('inf')))

ems Output Debug Console Terminal Ports
C:\Users\dines\Downloads> c;; cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscod
ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '57750' '--' 'c:\Users\dines\Downloads\exp_11.py'
--
ner? True
C:\Users\dines\Downloads> c;; cd 'c:\Users\dines\Downloads'; & 'C:\Users\dines\.local\bin\python3.14.exe' 'c:\Users\dines\.vscod
ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '63906' '--' 'c:\Users\dines\Downloads\exp_11.py'
imal value: 5
C:\Users\dines\Downloads>
```